

Mathematical Methods for Finance

Module 1 – Metric Spaces and Convergence

Lesson 3: Metric spaces, convergence and completeness

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3-hour lesson structure: intuition, theory, examples, exercises, and exam-style practice

Learning outcomes for Lesson 3

By the end of this lesson, students should be able to:

- ① verify whether a proposed formula is a metric;
- ② describe open balls, open sets, closed sets, interior and closure;
- ③ decide convergence of sequences in different metric spaces;
- ④ recognize Cauchy sequences and complete spaces;
- ⑤ use simple counterexamples to show why hypotheses matter.

Exam relevance

Metric spaces and convergence are not isolated definitions. They are the language used later for norms, function spaces, Banach spaces and Hilbert spaces.

How today is organised

14:30–15:20

Concepts and intuition

Distance and topology.

15:35–16:25

Theory and examples

Convergence, Cauchy sequences and completeness.

16:40–17:30

Exercises

Metric checks, convergence problems and counterexamples.

Module 1

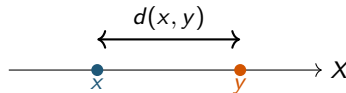
Block 1: distance and topology

Why start with metric spaces?

Intuition: The common idea

A metric space is a set where it makes sense to say that two objects are close.

- numbers can be close;
- vectors can be close;
- functions can be close;
- probability distributions can be close;
- approximations can be close to a target.



Metric spaces: definition

Definition: Metric

Let X be a set. A function $d : X \times X \rightarrow [0, +\infty)$ is a metric if, for all $x, y, z \in X$,

$$d(x, y) = 0 \iff x = y, \quad d(x, y) = d(y, x),$$

$$d(x, z) \leq d(x, y) + d(y, z).$$

The pair (X, d) is called a metric space.

Intuition: Interpretation

A metric is an abstract rule for measuring distance. It does not require coordinates or derivatives.

The four metric axioms

Identity

$d(x, y) = 0$ only if $x = y$.
Different points cannot have zero distance.

Symmetry

$d(x, y) = d(y, x)$.
Distance does not depend on direction.

Triangle

$d(x, z) \leq d(x, y) + d(y, z)$.
Going through y cannot improve the direct route.

Non-Negativity

$d(x, y) \geq 0, \forall x, y$

Exercise: Quick check ★

Which axiom fails for $d(x, y) = x - y$ on $\mathbb{R}$?

First examples of metrics

Example: Standard metrics

- ① On $\mathbb{R}$: $d(x, y) = |x - y|$.
- ② On $\mathbb{R}^n$: $d_2(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^2\right)^{1/2}$.
- ③ On any set X : $d(x, y) = 0$ if $x = y$, and $d(x, y) = 1$ if $x \neq y$.
- ④ On $C([a, b])$: $d_\infty(f, g) = \sup_{t \in [a, b]} |f(t) - g(t)|$.

Exercise: Recognition ★

For each example, identify what the points of the space are: numbers, vectors, symbols or functions.

Counterexamples to metric axioms

Counterexample: Not every distance-like formula is a metric

- ① $d(x, y) = (x - y)^2$ on $\mathbb{R}$ is nonnegative and symmetric, but fails the triangle inequality.
- ② $d(x, y) = |x^2 - y^2|$ on $\mathbb{R}$ fails identity: $d(1, -1) = 0$ but $1 \neq -1$.
- ③ $d(x, y) = |x - y| / (1 + |x - y|)$ is bounded, but boundedness is not a problem.

Exercise: Exam-style ★★

Show explicitly that $(x - y)^2$ fails the triangle inequality.

Solution: squared distance is not a metric

Proof idea: A single counterexample

Choose $x = 0$, $y = 1$, $z = 2$. Then

$$d(0, 2) = (0 - 2)^2 = 4.$$

But

$$d(0, 1) + d(1, 2) = 1^2 + 1^2 = 2.$$

The triangle inequality would require $4 \leq 2$, which is false.

Lesson

To disprove a universal property, one counterexample is enough.

Metrics generated by transformations

Theorem: A useful trick

If (X, d) is a metric space and $\varphi : [0, +\infty) \rightarrow [0, +\infty)$ is increasing, $\varphi(0) = 0$, and

$$\varphi(a + b) \leq \varphi(a) + \varphi(b),$$

then $\rho(x, y) = \varphi(d(x, y))$ is a metric.

Example: Bounded metric

With $\varphi(t) = t/(1 + t)$,

$$\rho(x, y) = \frac{d(x, y)}{1 + d(x, y)}$$

is a bounded metric.

Open balls

Definition: Open ball

In a metric space (X, d) , the open ball centered at $x \in X$ with radius $r > 0$ is

$$B(x, r) = \{y \in X : d(x, y) < r\}.$$

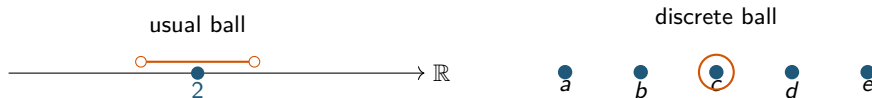
Example: Usual metric

In $\mathbb{R}$, $B(2, 0.5) = (1.5, 2.5)$.

Example: Discrete metric

If $0 < r \leq 1$, $B(x, r) = \{x\}$. If $r > 1$, $B(x, r) = X$.

Visual: balls in different metrics



Intuition: Same word, different geometry

An open ball is defined by a formula, but its shape depends completely on the metric.

Open and closed sets

Definition: Open set

A set $A \subset X$ is open if for every $x \in A$ there exists $r > 0$ such that $B(x, r) \subset A$.

Definition: Closed set

A set $F \subset X$ is closed if its complement $X \setminus F$ is open.

Example: On the real line

$(0, 1)$ is open; $[0, 1]$ is closed; $[0, 1)$ is neither open nor closed in $\mathbb{R}$.

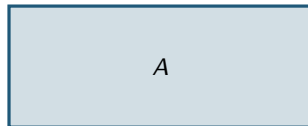
Interior, closure and boundary

Definition: Interior

A° is the set of points of A that contain a small ball entirely inside A .

Definition: Closure

$\bar{A}$ is the set of points that can be reached as limits of sequences from A .



boundary included in closure

Exercise: Check ★★

For $A = (0, 1)$ in $\mathbb{R}$, find A° , $\bar{A}$ and the boundary.

Solution: interval geometry

Worked example: $A = (0, 1)$ in $\mathbb{R}$

- Interior: every point of $(0, 1)$ has a small interval around it contained in $(0, 1)$, so $A^\circ = (0, 1)$.
- Closure: points 0 and 1 are limits of sequences from $(0, 1)$, so $\overline{A} = [0, 1]$.
- Boundary: $\overline{A} \setminus A^\circ = \{0, 1\}$.

Exam habit

For an interval, always check the endpoints separately.

Sequential characterization of closed sets

Theorem: Closed sets and limits

In a metric space, a set F is closed if and only if whenever $x_n \in F$ and $x_n \rightarrow x$, then $x \in F$.

Proof idea: Why this is useful

To prove that a set is closed, take a convergent sequence inside the set and show that its limit is still inside.

Exercise: Quick example ★★

Show that $[0, +\infty)$ is closed in $\mathbb{R}$ using sequences.

Continuous functions in metric spaces

Definition: Continuity

A map $f : (X, d_X) \rightarrow (Y, d_Y)$ is continuous at x if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d_X(x, z) < \delta \implies d_Y(f(x), f(z)) < \varepsilon.$$

Intuition: Meaning

Small changes in the input produce small changes in the output.

Theorem: Sequential criterion

In metric spaces, f is continuous at x iff $x_n \rightarrow x$ implies $f(x_n) \rightarrow f(x)$.

Block 1 mini-workshop

Exercise: Exercise A ★

Describe all open balls for the discrete metric when $r = 1/2$, $r = 1$, and $r = 2$.

Exercise: Exercise B ★★

On $\mathbb{R}$, decide whether $(-\infty, 2]$ is open, closed, both or neither.

Exercise: Exercise C ★★★

Let $d(x, y) = |x^3 - y^3|$ on $\mathbb{R}$. Prove that d is a metric.

Module 1

Block 2: convergence and completeness

What is a sequence?

Definition: Sequence

Let X be a set. A **sequence in X** is a function

$$x : \mathbb{N} \rightarrow X.$$

For each $n \in \mathbb{N}$, we write

$$x_n = x(n).$$

The sequence is denoted by

$$(x_n)_{n \in \mathbb{N}} \quad \text{or simply} \quad (x_n).$$

Intuition: An ordered list

A sequence can be viewed as an ordered list of elements of X :

$$x_1, x_2, x_3, \dots$$

The index n tells us which element of the list we are considering.

First examples of sequences

Example: $x_n = 1/n$

Consider the sequence

$$x_n = \frac{1}{n}.$$

Its first terms are

$$1, \quad \frac{1}{2}, \quad \frac{1}{3}, \quad \frac{1}{4}, \quad \dots$$

Here $X = \mathbb{R}$.

Example: $x_n = (-1)^n$

Consider the sequence

$$x_n = (-1)^n.$$

Its first terms are

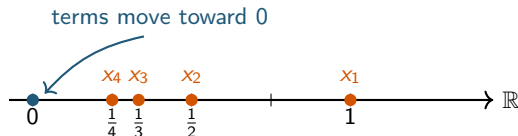
$$-1, \quad 1, \quad -1, \quad 1, \quad \dots$$

Again, $X = \mathbb{R}$.

Intuition: Different behaviours

A sequence does not have to move steadily in one direction or become constant. Its terms may approach a point, oscillate, or behave in more complicated ways.

Visual: a sequence on the real line



Example: $x_n = \frac{1}{n}$

The terms

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

can be represented as points on the real line.

Intuition: What we should notice

As n increases, the terms get closer and closer to 0. This is the basic picture behind convergence.

Sequences are not only sequences of numbers

Example: A sequence of vectors

In $\mathbb{R}^2$, define

$$x_n = \begin{pmatrix} 1/n \\ 1 - 1/n \end{pmatrix}.$$

Then

$$x_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}, \quad x_3 = \begin{pmatrix} 1/3 \\ 2/3 \end{pmatrix}, \quad \dots$$

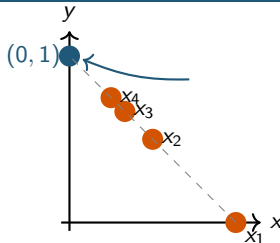
Intuition: The important point

The elements x_n can be numbers, vectors, functions, probability distributions, or other mathematical objects.

What matters is that

$$x_n \in X \quad \text{for every } n.$$

Visual: a sequence of vectors in $\mathbb{R}^2$



Example: Example: vectors

Consider

$$x_n = \begin{pmatrix} 1/n \\ 1 - 1/n \end{pmatrix}.$$

This is a sequence in $\mathbb{R}^2$.

Intuition: Geometric meaning

The terms are points in the plane, and they move toward the point $(0, 1)$. So a sequence may consist of vectors, not only real numbers.

From sequences to convergence

The question

Given a sequence $x_1, x_2, x_3, \dots$ in a metric space (X, d) , what happens to x_n as n becomes large?

Example: A first intuition

Consider

$$x_n = \frac{1}{n}.$$

Then

$$1, \quad \frac{1}{2}, \quad \frac{1}{3}, \quad \frac{1}{4}, \quad \frac{1}{5}, \quad \dots$$

and the terms appear to get closer and closer to 0.

Intuition: Next question

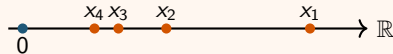
How can we translate

“ x_n gets arbitrarily close to x ”

into a precise mathematical statement?

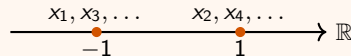
Visual: converging versus non-converging sequences

Example: Converging: $x_n = \frac{1}{n}$



The terms approach a single point: 0.

Example: Not converging: $x_n = (-1)^n$



The terms keep jumping between two points.

Intuition: Main lesson

A sequence converges only if its terms approach one specific point. If they keep oscillating, convergence fails.

Convergence in metric spaces

Definition: Convergence

Let (X, d) be a metric space. A sequence (x_n) converges to $x \in X$ if

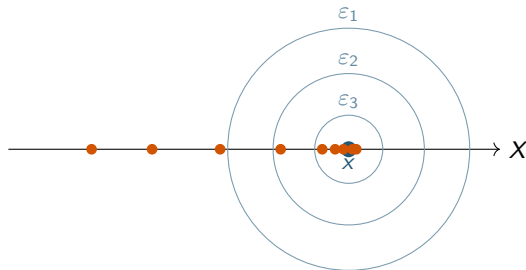
$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ such that } n \geq N \Rightarrow d(x_n, x) < \varepsilon.$$

We write $x_n \rightarrow x$.

Intuition: Meaning

Eventually, all terms of the sequence fall inside every ball centered at the limit.

Visual: convergence as shrinking balls



Reading the picture

For a large enough index n , the point x_n must remain in each prescribed ball around x .

Examples of convergence

Example: Usual metric

In $\mathbb{R}$, $x_n = 1/n$ satisfies $x_n \rightarrow 0$.

Example: Discrete metric

A sequence converges iff it is eventually constant.

Example: Uniform metric

On $C([0, 1])$, $f_n(t) = t^n$ does **not** converge uniformly to a continuous function.

Counterexample: Pointwise trap

Pointwise convergence is not the same as convergence in d_∞ .

Solution: convergence in the discrete metric

Proof idea: Characterization

In the discrete metric, $d(x_n, x) < 1/2$ implies $x_n = x$. If $x_n \rightarrow x$, choose $\varepsilon = 1/2$. Then there is N such that $n \geq N$ gives $d(x_n, x) < 1/2$. Hence $x_n = x$ eventually.

Theorem: Conclusion

A sequence converges in the discrete metric if and only if it is eventually constant.

Uniqueness of limits

Theorem: Uniqueness

In a metric space, if $x_n \rightarrow x$ and $x_n \rightarrow y$, then $x = y$.

Proof idea: Triangle inequality argument

For every n ,

$$d(x, y) \leq d(x, x_n) + d(x_n, y).$$

Let $n \rightarrow \infty$. Both terms on the right go to 0, hence $d(x, y) = 0$, so $x = y$.

Why metric spaces are convenient

The identity axiom makes limits unambiguous.

Cauchy sequences

Definition: Cauchy sequence

A sequence (x_n) in a metric space (X, d) is Cauchy if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ such that } m, n \geq N \Rightarrow d(x_n, x_m) < \varepsilon.$$

Intuition: Meaning

The terms eventually become close to each other, even before we know whether a limit exists.

Exercise: Check ★

Every convergent sequence is Cauchy. Prove it using the triangle inequality.

Solution: convergent implies Cauchy

Proof idea: Standard estimate

Assume $x_n \rightarrow x$. Given $\varepsilon > 0$, choose N such that for $n \geq N$,

$$d(x_n, x) < \varepsilon/2.$$

Then for $m, n \geq N$,

$$d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Thus (x_n) is Cauchy.

Completeness

Definition: Complete metric space

A metric space (X, d) is complete if every Cauchy sequence in X converges to a limit in X .

Intuition: Practical meaning

Completeness says that if approximations stabilize, their limit still belongs to the space where we are working.

Example: Examples

$\mathbb{R}$ with the usual metric is complete. $\mathbb{Q}$ with the usual metric is not complete.

Why $\mathbb{Q}$ is not complete

Counterexample: A Cauchy sequence without rational limit

There are rational sequences that converge in $\mathbb{R}$ to $\sqrt{2}$, for example decimal truncations

$$1, 1.4, 1.41, 1.414, 1.4142, \dots$$

They are Cauchy in $\mathbb{Q}$, but their limit is not in $\mathbb{Q}$.

The lesson

A sequence can be internally stabilizing while its natural limit falls outside the chosen space.

Completeness of closed subsets

Theorem: Closed subset of a complete space

If (X, d) is complete and $F \subset X$ is closed, then F is complete with the restricted metric.

Proof idea: Proof idea

Let (x_n) be Cauchy in F . It is also Cauchy in X , so it converges to some $x \in X$. Since F is closed and $x_n \in F$, we get $x \in F$.

Counterexample: Do not remove closedness

$(0, 1)$ is not complete as a subspace of $\mathbb{R}$: $x_n = 1/n$ is Cauchy but converges to $0 \notin (0, 1)$.

Uniform metric on continuous functions

Definition: Uniform distance

For $f, g \in C([a, b])$,

$$d_{\infty}(f, g) = \sup_{t \in [a, b]} |f(t) - g(t)|.$$

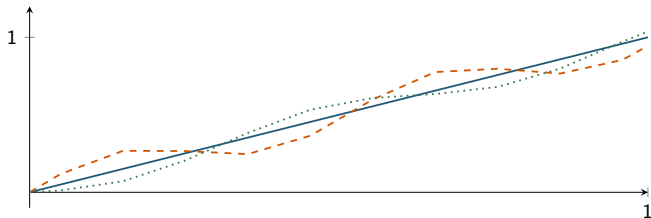
Theorem: Completeness

The space $C([a, b])$ with d_{∞} is complete.

Intuition: Reason

A uniformly Cauchy sequence of continuous functions has a uniform limit, and uniform limits of continuous functions are continuous.

Visual: uniform convergence



Intuition

Uniform convergence controls the whole graph at once, not just one point at a time.

Pointwise versus uniform convergence

Counterexample: Classical trap

Let $f_n(t) = t^n$ on $[0, 1]$. Then $f_n(t) \rightarrow 0$ for $0 \leq t < 1$, and $f_n(1) = 1$. The pointwise limit is

$$f(t) = \begin{cases} 0, & 0 \leq t < 1, \\ 1, & t = 1. \end{cases}$$

This limit is not continuous, hence convergence cannot be uniform in $C([0, 1])$.

Exercise: Exam-style ★★

Explain in words why this example shows that pointwise convergence is weaker than uniform convergence.

Block 2 mini-workshop

Exercise: Exercise A ★

Prove directly that $x_n = 1/n$ converges to 0 in $\mathbb{R}$.

Exercise: Exercise B ★★

Show that $x_n = n/(n+1)$ is Cauchy in $\mathbb{R}$ and find its limit.

Exercise: Exercise C ★★★

Give a Cauchy sequence in $(0, 1)$ that has no limit in $(0, 1)$.

Module 1

Block 3: guided exercises and exam training

Exam strategy for metric problems

Four-step template

- ① Write the relevant definition.
- ② Check the hypotheses one by one.
- ③ Use a theorem only after checking its assumptions.
- ④ If something is false, give explicit numbers or an explicit sequence.

Counterexample: Most common error

Students often say “it is obvious”. In an exam, replace this by a short verification.

Guided exercise 1: product metric

Exercise: Problem ★★

Let (X, d_X) and (Y, d_Y) be metric spaces. Define

$$D((x, y), (x', y')) = \max\{d_X(x, x'), d_Y(y, y')\}.$$

Show that D is a metric on $X \times Y$.

Plan

Check nonnegativity, identity, symmetry and triangle inequality. For the triangle inequality, use the two component triangle inequalities and then take the maximum.

Solution 1: product metric

Proof idea: Detailed steps

Nonnegativity and symmetry are inherited from d_X and d_Y . If $D((x, y), (x', y')) = 0$, both component distances are zero, hence $x = x'$ and $y = y'$. For the triangle inequality, for points p, q, r ,

$$d_X(p_X, r_X) \leq d_X(p_X, q_X) + d_X(q_X, r_X)$$

and similarly for Y . Therefore

$$D(p, r) \leq D(p, q) + D(q, r).$$

Guided exercise 2: closedness by sequences

Exercise: Problem ★★

Prove that the set

$$F = \{x \in \mathbb{R} : x \geq 3\}$$

is closed in $\mathbb{R}$.

Plan

Use the sequential characterization: take $x_n \in F$ and $x_n \rightarrow x$. Show $x \in F$.

Solution 2: closed ray

Proof idea: Detailed steps

Let $x_n \in F$ and assume $x_n \rightarrow x$ in $\mathbb{R}$. Since $x_n \geq 3$ for all n , suppose by contradiction that $x < 3$. Then $\varepsilon = (3 - x)/2 > 0$. For large n , $|x_n - x| < \varepsilon$, hence

$$x_n < x + \varepsilon = x + \frac{3 - x}{2} = \frac{x + 3}{2} < 3,$$

contradicting $x_n \geq 3$. Thus $x \geq 3$, so $x \in F$.

Guided exercise 3: non-complete interval

Exercise: Problem ★★

Show that $(0, 1)$ with the usual metric is not complete.

Plan

Find a Cauchy sequence entirely inside $(0, 1)$ whose limit in $\mathbb{R}$ does not belong to $(0, 1)$.

Example: Candidate

Use $x_n = 1/n$.

Solution 3: non-complete interval

Proof idea: Detailed steps

For every $n \geq 2$, $x_n = 1/n \in (0, 1)$. The sequence converges to 0 in $\mathbb{R}$, hence it is Cauchy. However, $0 \notin (0, 1)$. Therefore the sequence is Cauchy in the space but has no limit in the space.

Theorem: Conclusion

$(0, 1)$ is not complete.

Independent exercise set

Exercise: Exercise 1 ★

Let $d(x, y) = |x - y|$. Verify all metric axioms on $\mathbb{R}$.

Exercise: Exercise 2 ★★

Let $d(x, y) = |x^2 - y^2|$. Find the axiom that fails.

Exercise: Exercise 3 ★★★

Let (x_n) be Cauchy and suppose a subsequence converges to x . Prove that $x_n \rightarrow x$.

Solution sketch: Cauchy plus convergent subsequence

Proof idea: Key estimate

Let $x_{n_k} \rightarrow x$ and (x_n) be Cauchy. Given $\varepsilon > 0$, choose N_1 such that $m, n \geq N_1$ gives $d(x_m, x_n) < \varepsilon/2$. Choose k with $n_k \geq N_1$ and $d(x_{n_k}, x) < \varepsilon/2$. Then for $n \geq N_1$,

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \varepsilon.$$

Hence $x_n \rightarrow x$.

End-of-lesson checklist

Theorem: You should know

- metric axioms;
- open balls and open sets;
- convergence and Cauchy sequences;
- complete metric space;
- closed subset of a complete space.

Counterexample: You should avoid

- confusing pointwise and uniform convergence;
- assuming every Cauchy sequence has a limit;
- forgetting that the limit must belong to the space.

After-class work

Theorem: Core exercises

Exercise pack: 001, 002, 005, 008.

Example: Recommended

Exercise pack: 003, 006, 007, 011.

Advanced

Exercise pack: 018, 019, 020.

Exercise: Study method

For each exercise: write the relevant definition, check all hypotheses, then compute or build a counterexample.

Bridge to Lesson 4

Next question

Completeness tells us that Cauchy sequences converge. Compactness tells us that from many points we can extract convergent subsequences.

Intuition: Guiding image

A compact set is a space where sequences cannot escape.

Exercise: Prepare

Review closed and bounded intervals in $\mathbb{R}$, and think about why $[0, 1]$ behaves differently from $(0, 1)$.